Higher National Diploma in Information Technology
First Year, Second Semester Examination -2021
HNDIT2042- Data Communication and Computer Networks

Question 01

i.) Briefly explain Internet, Intranet and Extranet

ii) Explain 2-tier Client/Server model

iii) Explain benefits of 3-tier over 2-tier Client/Server model

iv) Write down four advantages of Leased Line

v) Compare and Contrast Fiber cable and Cat 6 Cable

Question 02

i.) Briefly explain three type of wireless media

ii.) Write four advantages of the Wireless Network

iii.) ABC company need to block facebook in the LAN. What solution you are going to propose?

iv) What is IoT

v) Write five loT Technologies and Trends

vi) Write down Two Cloud Business Applications

vii) Write down two Cloud Database Systems

Question 03

i.) List the Layers in TCP/IP reference Model

ii.) Write down the functions of any 2 Layers given in part (i) above.

iii.) Briefly explain the following protocols

a.) HTTPS

b.) ICMP

c.) SSH

d.) IMAP

iv.) Write Down suitable protocols for the following cases

a.) Hosts in a LAN are configured as automatically assign IP.

b.) Saman Download YouTube video using Internet

c.) A Lecturer conducting lectures using zoom

v.) Explain the functions of a Router

vi.) Explain the functions of a Switch

Question 04

i. Write down three classes of IPV4 and Private IP ranges of each classes.

ii. ABC company has three departments IT, HR, and Accounts. IT Department has 14 hosts, HR department has 28 hosts and Account Department has 90 hosts. ABC company need to divided three subnets. Fill in the blanks following tables. IP address gives as 192.168.1.1/24.

	IT Department	HR Department	Accounts Department
Subnet Marks	(a)……..	255.255.255.224	……..(c)
Uncast IP	192.168.1.0	(b)…….	……..(d)
Start IP	192.168.1.1	192.168.1.33	192.168.1.129
Last IP	192.168.1.30	192.168.1.62	……..(e)
Broad Cast IP	192.168.1.31	192.168.1.63	192.168.1.255
Number of Host	……..(f)	…….(j)	……..(g)
Number of IPs	32	…….(k)	…….(h)
CIDR	……..(m)	……..(l)	25

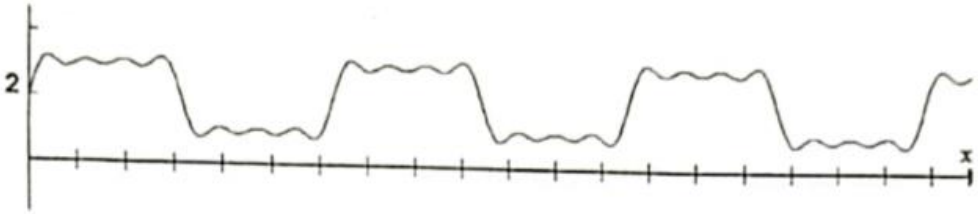
iii. Write a command to view computer logical address and physical address

iv. How to check whether Computers A and B are properly connected in the Network?

Assume IP address of A and B are 192.168.0.1 and 192.168.0.2

Question 05

i. Consider the following Periodic Composite Signal.



It can be mathematically represented as follows.

S(t) = 2 + 4/ π[sin x + 1/3sin3x + 1/5 sin 5x + 1/7 sin 7x]

a.) Draw the four frequency signal and one DC signal, (Hint-it can be decomposed using S (t) equations).

b.) Draw the Frequency Domain Plot for the above signals

ii.) Write down three computer security objectives

iii.) Explain Threats, Vulnerabilities, and Impact on computer security

Question 06

i.) Write down The Three Elements of Access Control
ii.) Explain User Account Control (UAC)
iii.) Write down three Logical access controls
iv.) Name three qualities that a good password should have
v.) Imagine you are a computer network administrator of an international company.

What practice should be followed for the security of his computer network? Write at least four.

vi.) Cyber-attacks are a persistent challenge that the world has yet to effectively prevent. The only solution is for everyone to adopt good cyber behavior, but unfortunately, this ideal is not easily achieved. What is your opinion on this matter?